

Seminar

Separation of Variables

A Formal Approach

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Reference

Trench, *Elementary Differential Equations*
Chapter 2, Section 2.2

Separable First-Order Ordinary Differential Equations

A general first-order ordinary differential equation can be written as

$$y' = F(x, y),$$

where

- ▶ x is the independent variable,
- ▶ $y = y(x)$ is the dependent variable,
- ▶ $y' = \frac{dy}{dx}$.

In this seminar, we will focus on the special class of equations for which $F(x, y)$ can be written as a product of a function of x and a function of y :

$$y' = f(x)g(y).$$

Assumptions

We assume that the functions involved are sufficiently well behaved (for example, continuous on the intervals under consideration), so that the required antiderivatives exist.

On an interval where $g(y) \neq 0$, define

$$h(y) = \frac{1}{g(y)}.$$

Then the equation can be written as

$$h(y) \frac{dy}{dx} = f(x).$$

This will be the form of the equation that we study.

A Different View of $\frac{dy}{dx}$

Recall that differentiation can be viewed as an operator:

$$\frac{d}{dx} : y \longmapsto y'.$$

That is,

$$\frac{dy}{dx} \text{ means } \left(\frac{d}{dx} \right) [y].$$

So in

$$h(y)y' = f(x),$$

we may write

$$h(y) \left(\frac{d}{dx} \right) [y] = f(x).$$

In this viewpoint, $\frac{dy}{dx}$ is not simply a ratio of two real numbers.

So why are we allowed to “move” dx below?

The Familiar Method

Starting with

$$h(y) \frac{dy}{dx} = f(x),$$

we write

$$h(y) dy = f(x) dx.$$

Integrating both sides,

$$\int h(y) dy = \int f(x) dx.$$

If H and F are antiderivatives of h and f , respectively, then

$$H(y) = F(x) + C.$$

But was this a legitimate mathematical move?

A New Representation

We want to separate the variables without treating $\frac{dy}{dx}$ as an ordinary quotient.

Starting from

$$h(y) \frac{dy}{dx} = f(x),$$

our goal is to **keep the differential operator intact**.

We will therefore look for a suitable function H such that

$$h(y) \frac{dy}{dx} = \frac{d}{dx} [H(y)].$$

Thus, instead of viewing the equation as

$$h(y) \frac{dy}{dx} = f(x),$$

we will represent it as

$$\boxed{\frac{d}{dx} [H(y)] = f(x).}$$

The equation is now in a form we already know how to solve:

$$\frac{d}{dx} [\text{something}] = f(x).$$

The mathematics of separation will emerge from this change of representation.

Constructing the Function H

Recall that we want to represent

$$h(y) \frac{dy}{dx}$$

as the derivative of some function of y .

Let I be an interval on which h is continuous. Fix $y_0 \in I$, and define

$$H(y) = \int_{y_0}^y h(t) dt.$$

By the Fundamental Theorem of Calculus I,

$$\frac{d}{dy} H(y) = h(y).$$

Thus, H is precisely an antiderivative of h :

$$\int h(y) dy = H(y) + C.$$

So the definite integral gives us a natural function H whose derivative is exactly h .

The Chain Rule Reveals the Representation

Now consider $H(y(x))$, where $y = y(x)$.

By the Chain Rule,

$$\frac{d}{dx} H(y(x)) = \frac{dH(y)}{dy} \frac{dy}{dx}.$$

Since

$$\frac{dH}{dy} = h(y),$$

we obtain

$$\boxed{\frac{d}{dx} H(y(x)) = h(y) \frac{dy}{dx}.$$

Therefore, our original equation

$$h(y) \frac{dy}{dx} = f(x)$$

can be represented as

$$\boxed{\frac{d}{dx} H(y(x)) = f(x).}$$

This is exactly the representation we were looking for.

*Choosing $H(y) = \int_{y_0}^y h(t) dt$ makes the chain rule
do the “separation” for us.*

Recovering the Familiar Separation

We obtained

$$H(y(x)) = \int f(x) dx.$$

By definition of H ,

$$H(y(x)) = \int_{y_0}^{y(x)} h(t) dt.$$

Thus,

$$\int_{y_0}^{y(x)} h(t) dt = \int f(x) dx.$$

Removing the limits of the definite integral and absorbing the fixed lower limit y_0 into the constant, we may write

$$\int h(y) dy = \int f(x) dx + C.$$

This is precisely the familiar “separation of variables” form.

The separation was achieved through the Chain Rule, not by treating dy/dx as a quotient.

A General Problem-Solving Philosophy

Change in Representation

A mathematical problem may be difficult in one representation, but become simple after representing the same object differently.

In our problem,

$$h(y) \frac{dy}{dx} = f(x)$$

was not easy to manipulate while keeping the differential operator intact.

We changed its representation to

$$\frac{d}{dx}H(y(x)) = f(x),$$

which was facilitated by the Chain Rule and the Fundamental Theorem of Calculus.

*The problem did not change.
Our representation of the problem did.*